**Resident Evil 4 .FCV Animation Format**

LegalSoul

NEW MEMBER

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Resident Evil 4 .FCV Animation Format Jul 6, 2024 at 11:54pm MrCurious and Biohazard4X like this

**Post by LegalSoul on Jul 6, 2024 at 11:54pm**

Hi all!

Apologies for being totally inactive - have too many things happening in life and had to abandon this lovely modding forum 😢

I was happy to see this thread and to find out that this FCV thing is still live!

Even though I can't dedicate time for this research myself, I shared the source code of my **SMDtoFCV** tool with MrCurious . The tool has its own tutorial here - residentevilmodding-boards.net/thread/5156/creating-animation-smdtofcv-converter-tutorial

Please note, as I mentioned in that tutorial of mine and as is said in this thread - my tool creates UNOPTIMIZED large fcv files and writes all keys even if there's no need for those keys. I didn't have enough resources and time to come up with a better tool, unfortunately. I also should note that the code that I shared is pretty bad - it was a long long time ago and was one of my first c++ coding attempts 😅

Anyway, please contact MrCurious if you wish to contribute to the creation of a proper FCV importer/exporter that we deserve 🙂

See you!

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Posts: 126

Last Edit: Jul 7, 2024 at 12:11am by LegalSoul

Resident Evil 4 .FCV Animation Format Jul 8, 2024 at 2:23pm LegalSoul, JADERLINK, and 1 more like this

**Post by Biohazard4X on Jul 8, 2024 at 2:23pm**

Nice to see ya around LegalSoul even if its one post for now haha. Thanks to the release of the PS2 debug it was possible to reverse engineer some of the code in the game thanks to zatarita and mariokart64n , we were able to fully document the FCV with everything contained there filling in some blanks while making some new mysteries(I'll get to that in a minute). The analysis below is based on that work and findings. Thanks for paving the way pioneer! Though good news indeed I'm afraid a long war must begin, testing and experimenting with the newly found documentation of the FCV some new nodes have been discovered and theirs still no viable way to really debug this file. You'l tool is kinda locked in place(but having the source code is great), we have Mariokart64n FCV script for 3ds max which is still mainly a debugging tool but CAN be used to create new animations but its need improved on and Mariokart64n doesn't really mess with RE4 anymore, and as a person who lacks any knowledge of programing or even the languages of programs due to not having time anymore but thats not the point, the information is out there.

As of 8/21/2024

Anyhow heres the newly documented format of the FCV...

```
********************
Header
********************

Animation Length  :Byte int16                 // Frame Count of the Entire Animation
Node Count Int8   :Byte int8                  // Specifies how many Node IDs are used/updated
Node Type         :Byte int8                  // Determines what kind of Node is being used  #amount depends on "Node Count"
Data TypeEX Pull :Byte int8                   // Determines how the data is store and IK pull #amount depends on "Node Count"
Node IDs  :Byte int8     // ID of Node; correlates to bone ID in Mesh #amount depends on "Node Count"
File Size         :Byte uint32      // Size of the File, can work nulled
Padding           :Array of Bytes   // Padding, To keep header in 16byte alignment, If needed
Block Data_address :Byte uint32               // Pointers for Nodes in Sequence of Appearance in Header here
Block Data        :Mix array                  // Keyframe Data for Nodes


***The Two Bytes that are conjointly are Node Type and Datatype. It will be explained below.***
*** These 2 bytes are a packed structure and represent 1 node, in their respective sequence.***
********************
Block data part
********************


Frame Count X :Byte int16       // How many keyframes the node will have
Frame Shift X :Byte int16       // Keyframe the Node will update on
Frame Data X  :Array of bytes // Explained Below, Specified by Data_type


Frame Count Y :Byte int16       // How many keyframes the node will have
Frame Shift Y :Byte int16       // Keyframe the Node will update on
Frame Data Y  :Array of bytes // Explained Below, Specified by Data_type


Frame Count Z :Byte int16       // How many keyframes the node will have
Frame Shift Z :Byte int16       // Keyframe the Node will update on
Frame Data Z  :Array of bytes // Explained Below, Specified by Data_type



********************
Data Type Explained (Refer to examples below for more info)
********************
0x00: Float  ;  Float  ;  Float  ; 12byte = 4:4:4
0x10: Float  ;  16int  ;  16int  ;  8byte = 4:2:2
0x20: Float  ;  Bint   ;  Bint   ;  4byte = 4:1:1
0x40: 16int  ;  Float  ;  16int  ; 16byte = 2:4:4
0x60: 16int  ;  16int  ;  16int  ;  8byte = 2:2:2
0x80: Bint   ;  16int  ;  Bint   ;  4byte = 1:2:1
0x90: Bint   ;  16int  ;  16int  ;  8byte = 1:2:2
0xA0: Bint   ;  Bint   ;  Bint   ;  3byte = 1:1:1


********************
Examples of IK Pull (Relative/Local)
********************
0x00  : pLayers Left
0x01  : players Right
0x02  : players Front
0x03  : players Right
0x04  : Players Front
0x05  : players back
0x06  : Camera Type // Not Actually IK Pull; Specifies if the Node ID will be a Camera related node)


********************
Examples of Bone_Type (Bit Flags)
********************

TypeOx01:    Player Movement (Relative)    // Controller node to move the PC/NPC through the scene
TypeOx02:    Rotation        (Relative)    // FK Node
TypeOx04:    Translation     (Relative)    // IK Node
TypeOx08:    Scale           (Relative)    // Scaling Node
TypeOx10:    Rotation        (Absolute) ?
TypeOx20:    Translation     (Absolute) ?  // This would make sense as there is code for world transformations
TypeOx40:    Scale           (Absolute) ?  // would make sense with the existing pattern
TypeOx80:    Flip Bone   // Used for swapping between left and righted handed enemies, needs researched


********************
Camera Nodes
********************
Nodes cam be identified if its specified Data/IX pull ends with a 0x06
The known nodes are...

Node ID 00: Camera Position      // Position for Camera
Node ID 01: Camera Target        // Target Position for Camera
Node ID 02: Camera Roll // Roll Parameter for Camera
Node ID 03: Camera FOV           // Field of Vision
Node ID 04: Camera Speed         // Camera Speed; Can speed up and slow down camera, unsure how it works
Node ID 05: ? ??

********************
Examples of Block_data by Data_type
********************

$00: Float   ;  Float  ;  Float  ;


    4 bytes        4 bytes        4 byteso
  8E F3 2F C7     00 00 00 00     FE 13 23 44 = 3 sets of 32Bit Float = 12 bytes Total
  79 82 D6 C7     00 F8 05 44     2E 2F A7 43 = 3 sets of 32Bit Float = 12 bytes Total
KeyFrame Data    Spline In       Spline Out


$10: Float   ; 16int   ;  16int  ;


   4 bytes  2 Bytes 2 Bytes
 19 99 14 3F   00 00 E0 C0 = 8 bytes
  73 39 C9 3F    56 F8          68 40 = 8 bytes
KeyFrame Data   Spline In    Spline Out


Splines/Curves are used as HermIte splines.
********************
Data Types Explained
********************


0x30: Float  ; 16int  ; 16int
 4bytes   2Bytes  2Bytes
  IX/RX        Anim_Curves
 Data             Data


NOTES:
Involving the animation curves these can also be similar to curves involving image animation and ect
In 3ds max these are also called, Incoming and outgoing tangents
@MarioKart64n found out they were Hermite splines.
```

Also for anyone interested in the 3Ds max Script for the FCV made by MarioKart64n heres the link --> Download FCV Script for 3DsMax (Updated 8/21/2024)

Last Edit: Mar 31, 2025 at 8:00pm by Biohazard4X: Made Subtle Changes...

Biohazard4X

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coach

NEW MEMBER

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Resident Evil 4 .FCV Animation Format Aug 17, 2024 at 5:31am

**Post by coach on Aug 17, 2024 at 5:31am**

do you have a direct link to this tool? your link is dead

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Biohazard4X

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Resident Evil 4 .FCV Animation Format Aug 21, 2024 at 10:35pm

**Post by Biohazard4X on Aug 21, 2024 at 10:35pm**

Aug 17, 2024 at 5:31am coach said:

do you have a direct link to this tool? your link is dead

**Thanks** for bringing to my attention! Updated format a little bit for ease and updated Link with a more permanent link!

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DANIEL FERREIRA

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Resident Evil 4 .FCV Animation Format Feb 27, 2025 at 1:29pm

**Post by DANIEL FERREIRA on Feb 27, 2025 at 1:29pm**

Hello Greetings Biohazard4X i just interested in your .FCV editing and i asking for you:

if just extracting the em20.uda located in em folder and get the super dr. salvador FCV animation do you know the .FCV list

for each action and post this forum? or the all FCV list for each enemy it's too large?

You made it prolonged your l6s but not to escape to your inevitable death isn't

Posts: 58

Biohazard4X

NEW MEMBER

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Resident Evil 4 .FCV Animation Format Feb 27, 2025 at 2:42pm

**Post by Biohazard4X on Feb 27, 2025 at 2:42pm**

Feb 27, 2025 at 1:29pm DANIEL FERREIRA said:

Hello Greetings Biohazard4X i just interested in your .FCV editing and i asking for you:

if just extracting the em20.uda located in em folder and get the super dr. salvador FCV animation do you know the .FCV list

for each action and post this forum? or the all FCV list for each enemy it's too large?

Hello Thanks for the interest but I was hesitant to keep this thread dedicated to the technical side of the FCV and not what each FCV does or is, their are plenty of threads on that. Even if their isn't then that warrants making another thread dedicated your followine